Band-1 Hadamard Order 8948 – Miyamoto (Paley-II Seed H_2236 q=1117 + C_2238); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_2236 Paley-II PASS; C_2238 PASS; H_8948 Miyamoto PASS; H_8948 Paley-II SKIP (q=4473 not prime power); Miyamoto-only; skip 8788/8836 honest; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-21_Band1_Hadamard_8884_Miyamoto`; `2026-09-16_Band1_Hadamard_8644_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_8948_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 2236 and 8948 only (verified max|HHˆT-nI|=0), via Miyamoto with Paley-II seed H_2236 (q=1117) + C_2238 over GF(2237); 8948 Miyamoto-only NOT in SPK.** Honest skips: **8788** (GF(2197)=13ˆ3 unimplemented; `paley_core_prime_power` k=1,2 only); **8836** (seed Paley-II q=1103==3 mod4 fails codebase rule). Classical SPK for 8948: Paley-I **FAIL**s ($q = 8947 = 23 \cdot 389$ NOT prime); Paley-II **FAIL** ($q = 4473 = 3^2 \cdot 7 \cdot 71$ NOT prime power); Kronecker **FAIL**s (4×2237). Seed **2236** folds into SPK when Paley-II seed verifies; **8948** is Miyamoto-only (**NOT** folded into SPK). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **2220+8884+2160+2140+8564 / 1064+2128 / ...** retained; Miyamoto-only **8644/8516/8452/7988/... (+8948)** excluded.

Abstract

Finite constructions of orders **2236** and **8948**.

- H_{2236} = Paley-II ($q = 1117$) – preferred Miyamoto seed
- C_{2238} = Paley conference over GF(2237), $q = 2237$
- H_{8948} = Miyamoto(H_{2236}, C_{2238})
- Paley-II for 8948: **SKIP** ($q = 4473 = 3^2 \cdot 7 \cdot 71$ NOT prime power)
- Paley-I for 8948: **SKIP** ($q = 8947 = 23 \cdot 389$ NOT prime)
- Kronecker for 8948: **SKIP** ($4 \cdot 2237$ not admissible)
- Skips: **8788** (GF(13^3)); **8836** (seed q=1103==3)

SPK: Miyamoto-only NOT in SPK.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8948.py`

Results. overall_pass=True; units 168/168; wall 596.7425s; all constructed max|HH^T-nI|=0.

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Arithmetic

- Target $n = 8948 = 4 \cdot 2237$; 2237 prime $\equiv 1 \pmod 4$ -> Miyamoto-eligible.
- Need H_{2236} and C_{2238} (Paley conference over prime field $q = 2237$).
- Seed: Paley-II $q = 1117$ (preferred); Paley-I $q = 2235$ FAIL; Double FAIL; Kronecker 4×559 FAIL.
- SPK: Paley-I $q = 8947$ **FAIL**; Paley-II $q = 4473$ **FAIL** (not prime power); Kronecker **FAIL**.
- Prior optional skipped: 8788 (GF(13^3)); 8836 (seed ==3 mod4).

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_2236 Paley-II	Paley-II q=1117	0	5.9029
C_2238	Paley conf GF(2237)	0	7.9605
H_8948 Miyamoto	Miyamoto(H_2236, C_2238)	0	575.5955
H_8948 Paley-II	SKIP q=4473=3^2*71 NOT prime power	–	–
H_8948 Paley-I	SKIP q=8947=23*389	–	–
H_8948 Kronecker	SKIP 4*2237 not admissible	–	–

SPK closure note

Seeds include prior SPK through **8884** lineage (**2220+8884**; **2160**; **2140+8564**; **1064+2128**; . . . ; **NOT** 8644/8516/8452/. . . /+8948) + verified **2236**. Next constructive gap **2052** (OPEN).

Quantity	Value
8948 in SPK	**False** (Miyamoto-only)
2236 in SPK	**True** (Paley-II seed)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=2269->n=9076 PaleyI(q=2267) OR q=2273->n=9092 Kronecker(4x568) OR q=2281->n=9124 triage OR q=2293->n=9172 triage OR method-attack SPK gap 2052 OR Track B / Object 1 meter await